

Rules for writing hamza

1 Introduction

The rules of how to write hamza in the Arabic script are quite complex. The matter is further complicated by the limitations of metal and digital font technology. We attempt to give a comprehensive set of rules for writing hamza and some recommendations for typographers and typesetters.

This article is based on findings from the author's research. Any errors reported as issues at <https://github.com/adamiturabi/hamza-rules> will be greatly appreciated.

2 Scope

We treat only the orthography of Standard Arabic. The orthography of the Qurʾān (الرسم العثماني) is not discussed.

3 Seats of hamza

hamza is written in four different ways:

1. Seated on an 'alif: أ or إ
2. Seated on an wāw: و
3. Seated on an yā': ي
4. Unseated: ء

Here are some of notes about writing hamza in the above four methods:

- When unseated hamza comes between two letters that are joined, then it is written above the line that joins them, for example: كَهَاتِيءة *k̥haṭīʾah*. In this word, the yā' ي joins to the tā' *marbūṭah* ة.

As a special case, when unseated hamza comes between joined lām and 'alif (لا), then it is positioned between them thus: لاء. (In most cases, this is replaced with لّ as we will explain in the next point below.) And this is different from hamzah on the 'alif following the lām: لأ.

- When unseated hamza is followed by an 'alif: ء, the combination of hamza and 'alif is usually written as Ā as a convention. Examples: آمَنَ 'āmana, ظَمَّانَ ḍham'ān, شَنَّانَ shana'ān. However, when the 'alif is a suffix or part of a suffix, or the hamza is doubled, or there is an 'alif before the hamza then we will write ء, not Ā. Examples: شَيِّءَانِ shay'āni, سَاءَال sa''āl, قِرَاءَات qirā'āt.
- When hamza is seated on 'alif, if it has an *kasrah*, it is written below the 'alif: اِ. Otherwise, it is written above the 'alif: اَ, اُ, اِ.
- When hamza is seated on yā' ي the dots of the yā' are no longer written. Here's how it will appear in different positions:

Isolated	End	Middle	Beginning
ي	ي	ي	ي

Note that hamza is seated on yā' in the middle position ي is different from unseated hamza between two joining letters يـ.

So how do we know when to write hamza unseated and when seated? And how do we choose between its three different seats? There are a set of rules that we need to follow in order to correctly write hamza.

4 Rules for determining the seat of hamza

4.1 Without prefixes and suffixes

We will first learn how to determine the seat of hamzah for a word without any prefix or suffix.

Hamzah can occur in three positions in a word:

1. At the beginning of the word
2. In the middle of the word
3. At the end of the word

We will treat each of these positions below.

4.1.1 At the beginning of the word

When hamza occurs in the beginning of a word, then:

- a. If the hamza carries a long-ā vowel, it is written unseated followed by an 'alif and written as Ā, for example آمَنَ 'āmana.
- b. If the hamza carries any other vowel, it is written seated on an 'alif, and is marked with the appropriated vowel mark, for example أَسْلَمَ 'aslama, أُورِدُ 'uridu, اِسْلَامَ 'islām, اِيْمَانِ 'imān, اَوْحَدَ 'ūkhidha.

4.1.2 In the middle of the word

The most general case is when hamza is in the middle of a word.

Arabic has three short vowels, three long vowels, two diphthongs, and a *sukūn*. Each of these has an order of precedence and a hamza seat.

Precedence	Vowel	Seated hamzah
1.	<i>ī/ay</i>	ء
2.	<i>i</i>	ئ
3.	<i>ū/aw</i>	ء
4.	<i>u</i>	ؤ
5.	<i>ā</i>	ء
6.	<i>a</i>	أ
7.	◌ْ	ء

Main rule: Disregard any doubling mark ◌ْ and consider the vowel on the consonant before the hamza and the *shortened* vowel on the hamza itself. Determine which of the two vowels wins by being higher in precedence in the above table. The winning vowel's seat will be the seat of the hamza.

Sub-rule: If the main rule determines that hamza is to be seated on 'alif, and there is a long *ā* vowel on the hamza using an 'alif, then hamza shall be unseated. And the combination of ء will usually be written as ĩ.

Examples:

Word	Vowel on consonant before hamza	Shortened vowel on hamza	Winning vowel	Seated hamza
هَيَّاهُ <i>hay'ah</i>	<i>ay</i>	<i>a</i>	<i>ay</i>	ء
خَطِيَّاهُ <i>khaṭī'ah</i>	<i>ī</i>	<i>a</i>	<i>ī</i>	ء
اِسْتِيَّاسُ <i>'istī'ās</i>	<i>ī</i>	<i>a</i>	<i>ī</i>	ء (Exception: ء is not written as ĩ when the preceding vowel is <i>ī</i> .)
تَوَّامُ <i>taw'am</i>	<i>aw</i>	<i>a</i>	<i>aw</i>	ء
سَائِلُ <i>sā'il</i>	<i>ā</i>	<i>i</i>	<i>i</i>	ئ
تَسَاوُلُ <i>tasā'ul</i>	<i>ā</i>	<i>u</i>	<i>u</i>	ؤ
تَسَاوَلُ <i>tasā'ala</i>	<i>ā</i>	<i>a</i>	<i>ā</i>	ء
قِرَاءَاتُ <i>qirā'āt</i>	<i>ā</i>	<i>a</i>	<i>ā</i>	ء
نُوَّانُ <i>nū'ānun</i>	<i>ū</i>	<i>a</i>	<i>ū</i>	ء
مَسْوُولُ <i>mas'ul</i>	◌ْ	<i>u</i>	<i>u</i>	ؤ
تَرْئِيسُ <i>tar'is</i>	◌ْ	<i>i</i>	<i>i</i>	ئ
مِرَّاهُ <i>mir'ah</i>	◌ْ	<i>a</i>	<i>a</i>	ء (Using sub-rule.)

Word	Vowel on consonant before hamza	Shortened vowel on hamza	Winning vowel	Seated hamza
ظَمَّانَ <i>ḡhamʿān</i>	◌َ	<i>a</i>	<i>a</i>	ء (Using sub-rule.)
مَسْأَلَةٌ <i>masʿalah</i>	◌َ	<i>a</i>	<i>a</i>	أ
الْمَرْأَةُ <i>almarʿah</i>	◌َ	<i>a</i>	<i>a</i>	أ
بَيْتِ <i>biʿsa</i>	<i>i</i>	◌ِ	<i>i</i>	ئ
سُؤْلٍ <i>suʿl</i>	<i>u</i>	◌ِ	<i>u</i>	ؤ
كَاسٍ <i>kaʿs</i>	<i>a</i>	◌ِ	<i>a</i>	أ
سُؤْلٍ <i>suʿila</i>	<i>u</i>	<i>i</i>	<i>i</i>	ئ
يَيْسٍ <i>yaʿisa</i>	<i>a</i>	<i>i</i>	<i>i</i>	ئ
رَيْسٍ <i>raʿis</i>	<i>a</i>	<i>i</i>	<i>i</i>	ئ
سُؤَالٍ <i>suʿāl</i>	<i>u</i>	<i>a</i>	<i>u</i>	ؤ
رُؤُوسٍ <i>ruʿūs</i>	<i>u</i>	<i>u</i>	<i>u</i>	ؤ
لُؤْيٍ <i>luʿayy</i>	<i>u</i>	<i>a</i>	<i>u</i>	ؤ
شَنَانٍ <i>shanaʿān</i>	<i>a</i>	<i>a</i>	<i>a</i>	ء (Using sub-rule.)
سَأَلَ <i>saʿala</i>	<i>a</i>	<i>a</i>	<i>a</i>	أ
رَأَى <i>raʿā</i>	<i>a</i>	<i>a</i>	<i>a</i>	أ (Sub-rule doesn't apply because <i>ā</i> vowel at end represented by ئ, not ʿalif.)
رَأَسَ <i>raʿasa</i>	<i>a</i>	<i>a</i>	<i>a</i>	أ
يُرَئِسُ <i>yuraʿisu</i>	<i>a</i>	<i>i</i>	<i>i</i>	ئ
رُئِسَ <i>ruʿisa</i>	<i>u</i>	<i>i</i>	<i>i</i>	ئ
تَقُولُ <i>tafaʿul</i>	<i>a</i>	<i>u</i>	<i>u</i>	ؤ
سَأَلَ <i>saʿāl</i>	<i>a</i>	<i>a</i>	<i>a</i>	ء (Using sub-rule.)
لَأَلَ <i>laʿāl</i>	<i>a</i>	<i>a</i>	<i>a</i>	ء (Using sub-rule.)

4.1.3 At the end of the word

When hamza occurs at the end of a word, disregard the vowel on hamza itself, and consider only the vowel on preceding consonant. Plug it into the precedence table as above to determine the seat of hamza.

Word	Vowel on consonant before hamza	Seated hamza
دُعَاءُ <i>dueʿāu</i>	<i>ā</i>	ء
سُوءٌ <i>sūʿu</i>	<i>ū</i>	ء
جِيءَ <i>jīʿa</i>	<i>ī</i>	ء
دَوَّءَ <i>ḡawʿa</i>	<i>aw</i>	ء
شَيَّءَ <i>shayʿa</i>	<i>ay</i>	ء

Word	Vowel on consonant before hamza	Seated hamza
بُطْءُ <i>buṭʿu</i>	◌ْ	ء
عِبْءُ <i>ʿibʿu</i>	◌ْ	ء
شَطْءُ <i>shatʿu</i>	◌ْ	ء
يُهَدِّى <i>yuhaddiʿu</i>	i	ئ
سَيِّى <i>sayyiʿu</i>	i	ئ
بَطْءُ <i>baṭuʿa</i>	u	ؤ
يَهْدَا <i>yahdaʿu</i>	a	أ
مُبْتَدَا <i>mubtadaʿi</i>	a	إ

The exception to this rule is when the previous letter is a doubled *wāw* with an *ḍammaḥ*. In this case the hamza will again be unseated. Example تَبَوُّءُ *tabawwuʿu*.

Note also that مُبْتَدَا *mubtadaʿi* can be written with the hamza below the *ʿalif* because of the *i*-mark on the hamza. But it is also common to write it as مُبْتَدَا *mubtadaʿ*, especially when the hamza is unvoveled.

4.2 Prefixes and suffixes

4.2.1 Prefixes

If hamza is in the beginning of a word, adding a prefix to the word will not alter the writing of the hamza. Examples:

- ل + أُسْتَاذٍ = لِأُسْتَاذٍ
- اِنْ + آخِرَةٌ = الْآخِرَةُ

4.2.2 Suffixes

If hamza is at the end of a word, adding a suffix to the word can, in general, alter the writing of the hamza. Hamza is now, generally, treated as if it is in the middle of the word, and the rules for hamza in the middle of a word apply. Examples:

Word	Vowel on consonant before hamza	Shortened vowel on hamza	Winning vowel	Seated hamza
بَرِيءُونَ <i>bariʿūna</i>	<i>ī</i>	<i>u</i>	<i>ī</i>	ء
بَرِيءَانِ <i>bariʿāni</i>	<i>ī</i>	<i>a</i>	<i>ī</i>	ء
بَرِيءِينَ <i>bariʿīna</i>	<i>ī</i>	<i>i</i>	<i>ī</i>	ء

Word	Vowel on consonant before hamza	Shortened vowel on hamza	Winning vowel	Seated hamza
بَرِيْءَيْنِ <i>bari'ayni</i>	<i>ī</i>	<i>a</i>	<i>ī</i>	ء
شَيْءُهُ <i>shay'uhu</i>	<i>ay</i>	<i>u</i>	<i>ay</i>	ء
شَيْءَاهُ <i>shay'ahu</i>	<i>ay</i>	<i>a</i>	<i>ay</i>	ء
شَيْءِيهِ <i>shay'ihī</i>	<i>ay</i>	<i>i</i>	<i>ay</i>	ء
شَيْءَانِ <i>shay'āni</i>	<i>ay</i>	<i>a</i>	<i>ay</i>	ء
شَيْءَيْنِ <i>shay'ayni</i>	<i>ay</i>	<i>a</i>	<i>ay</i>	ء
مَجِيْءُهُ <i>majī'uhu</i>	<i>ī</i>	<i>u</i>	<i>ī</i>	ء
مَجِيْءَاهُ <i>majī'ahu</i>	<i>ī</i>	<i>a</i>	<i>ī</i>	ء
مَجِيْءِيهِ <i>majī'ihī</i>	<i>ī</i>	<i>i</i>	<i>ī</i>	ء
سُوْءِيهِ <i>sū'ihī</i>	<i>ū</i>	<i>i</i>	<i>i</i>	ئ
صَوْءِيهِ <i>ḍaw'ihī</i>	<i>aw</i>	<i>i</i>	<i>i</i>	ئ
سُوْءُهُ <i>sū'uhu</i>	<i>ū</i>	<i>u</i>	<i>ū</i>	ء
سُوْءَاهُ <i>sū'ahu</i>	<i>ū</i>	<i>a</i>	<i>ū</i>	ء
سُوْءَانِ <i>sū'āni</i>	<i>ū</i>	<i>a</i>	<i>ū</i>	ء
صَوْءَاهُ <i>ḍaw'ahu</i>	<i>aw</i>	<i>a</i>	<i>aw</i>	ء
صَوْءَانِ <i>ḍaw'āni</i>	<i>aw</i>	<i>a</i>	<i>aw</i>	ء
مُتَّكِيْنَ <i>muttaki'īna</i>	<i>i</i>	<i>i</i>	<i>i</i>	ئ
يُبَرِّئُوْنَ <i>yubarrī'ūna</i>	<i>i</i>	<i>u</i>	<i>i</i>	ئ
يُبَرِّئُوْنَ <i>yubarra'ūna</i>	<i>a</i>	<i>u</i>	<i>u</i>	ؤ

There are some exceptions:

- If the letter before the hamza has a *sukūn* and is not *wāw* or *yā'*, then the hamza will be written unseated. Examples:

- جُرْءَانِ *juz'āni*
- عِبْءَانِ *ʿib'āni*
- عِبْءَيْنِ *ʿib'ayni*
- بُطْءَاهُ *buṭ'ahu*
- بُطْءُهُ *buṭ'uhu*
- بُطْءِيهِ *buṭ'ihī*

- If the hamzah is after a long-*ū* vowel or an *aw* semi-vowel, and the hamzah does not have an *i*-mark, then it is written unseated. Examples:

– يَسُوْءُوْنَ *yasū'ūna*

This is actually an acceptable variant in the general case, as will discuss below in section @ref(hamza-variants)

(ان, ُ, ِ, and ٍ are suffixes.) Note that the combination ٰء is not written as ̣ when the 'alif is part of the suffix.

4.3 *tanwīn* on final hamza

tanwīn on a final hamza does not affect the writing of the hamza except in the case of *tanwīn al-fatḥ*. When writing *tanwīn al-fatḥ* on a hamza at the end of a word:

1. If there is an 'alif before a unseated hamza ٰ, then we don't add a silent 'alif when writing *tanwīn al-fatḥ*. For example ٰء becomes ٰء *dā'an*, not ٰء.
2. Otherwise, we add the silent 'alif after the hamza so that the hamza is now in the middle of the word with a suffix 'alif after it. We now pretend that the hamza has an *fatḥah* and that the 'alif after it is a long-ā vowel. Then we go through the rules for writing hamza in the middle of a word (given above) to determine how hamza will be written. We then write the *an*-mark on the hamza. Examples:

- مُبْتَدَأٌ becomes مُبْتَدَأًا, مُبْتَدَأًا, مُبْتَدَأًا
- مَلَجًا becomes مَلَجَاءًا, مَلَجَاءًا, مَلَجَاءًا
- جُزْءٌ becomes جُزْءًا, جُزْءًا, جُزْءًا
- شَيْءٌ becomes شَيْءًا, شَيْءًا, شَيْءًا
- سَبِيٌّ becomes سَبِيًّا, سَبِيًّا, سَبِيًّا
- تَبَوُّءٌ becomes تَبَوُّوءًا, تَبَوُّوءًا, تَبَوُّوءًا

4.4 Variants

There are some historical and regional variants to the above rules. The main one is when the letter before hamza has a *sukūn*, the hamza is generally written unseated. So with this variant, we write:

- مَسْئُولٌ instead of مَسْئُولًا
- أَسْئَلَةٌ instead of أَسْئَلَةً
- مَسْأَلَةٌ instead of مَسْأَلَةً

However, this rule appears to be not consistently followed. For example, *nash'ah* is generally always written نَشَاءَةٌ never نَشْءَةٌ.

A second variant is to avoid the repetition of vowel letters like و and ي. So they write:

1. رُوُوسٌ instead of رُوُوسًا.
2. رُؤُوسٌ instead of رُؤُوسًا.

5 Typographical limitations

Due to what appears to have been a limitation of typesetting technology in the days of typewriters, metal typography, and early digital typography, unseated hamza between two joining letters **ءـ** was usually written as seated on *yā*’ instead: **ئـ**. Because of this limitation we are now accustomed to seeing:

- شَيْئًا instead of شَيْءًا
- خَطِيئَةٌ instead of خَطِيءَةٌ
- هَيْئَةٌ instead of هَيْءَةٌ
- عَبَّائِينَ instead of عَبَّءِينَ

and similar variants.

These variants have pervaded to such a degree that many modern explanations on the rules of hamza orthography present the above as the correct way of writing, and modify their rules with exceptions to allow this writing.

Fortunately, advancements in digital font technology now allow us to revert back to the original rules. However, unfortunately, only very few computer fonts today actually implement this feature.

Two fonts, of which we are aware, that do allow the preferred orthography are:

1. Dr Khaled Hosny’s “Amiri”: <https://www.amirifont.org/>
2. DecoType “Naskh”: <https://www.decotype.com/oneliner/>

With these “hamza-safe fonts” you can always use U + 0621 “Arabic Letter Hamza” to type unseated hamza no matter whether it is between joining or non-joining letters. With most other fonts U + 0621 “Arabic Letter Hamza” will prevent the surrounding two letters from joining.

This issue has been discussed in detail by Thomas Milo in Unicode L2/14-109: <https://unicode.org/L2/L2014/14109-inline-chars.pdf>.

However, the problem remains that this is a font-specific hack. It seems that the Unicode has no official guidance in this regard.

Another solution, that works with most other fonts, is to fake the correct orthography using a combination of U + 0640 “Arabic Tatweel” and U + 0654 “Arabic hamza above” thus **ءـ** to achieve the appearance of unseated hamza between two joining letters. However, this will not work when hamza is between *lām* and *ʿalif* in the mandatory *lām-ʿalif* ligature **لا** (correct: **لا**, incorrect: **لا**). Such words are rare in Standard Arabic but do exist, e.g. **لآل** *laʿāl* meaning “pearl-seller”, and **مِلْءَا**. Perhaps the official solution ought to be that fonts absorb the tatweel and not let it affect the *lām-ʿalif* ligature when it is input between *lām* and *ʿalif*.

Beware that using tatweel in this manner may also affect the searching of characters in a digital document.